Edge Computing in Industrial Communication

TEAM-A5

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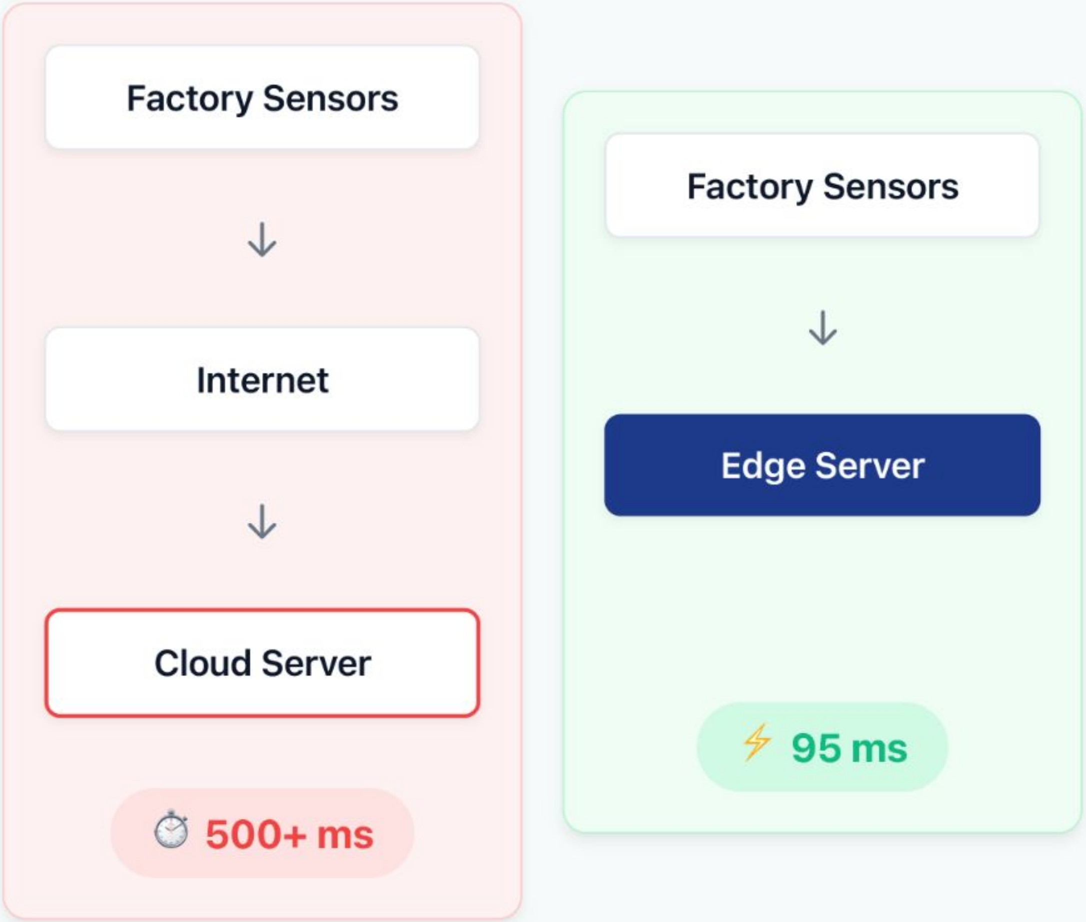
Why Do We Need Edge Computing?

The Problem:

- Billions of IoT devices generate data.
- Industrial systems need millisecond responses.
- Sending everything to the cloud causes delay.
- Delays can damage equipment and reduce safety.

Edge computing brings computation closer to the machines.



The Industrial Problem

Example: Robotic Welding Station

Cloud Processing

Weld defect detected

↓

Data sent to Cloud

↓

Response arrives

↓

Machine Damaged (Too Late)

Edge Processing

Weld defect detected

↓

Local Edge Node

↓

Immediate Shutdown

How Edge Computing Works

☁️ Cloud Layer

Long-term storage & Historical Analysis

↑ **Only important results go to cloud** ↑

⚙️ Edge Layer

Local processing, immediate actions

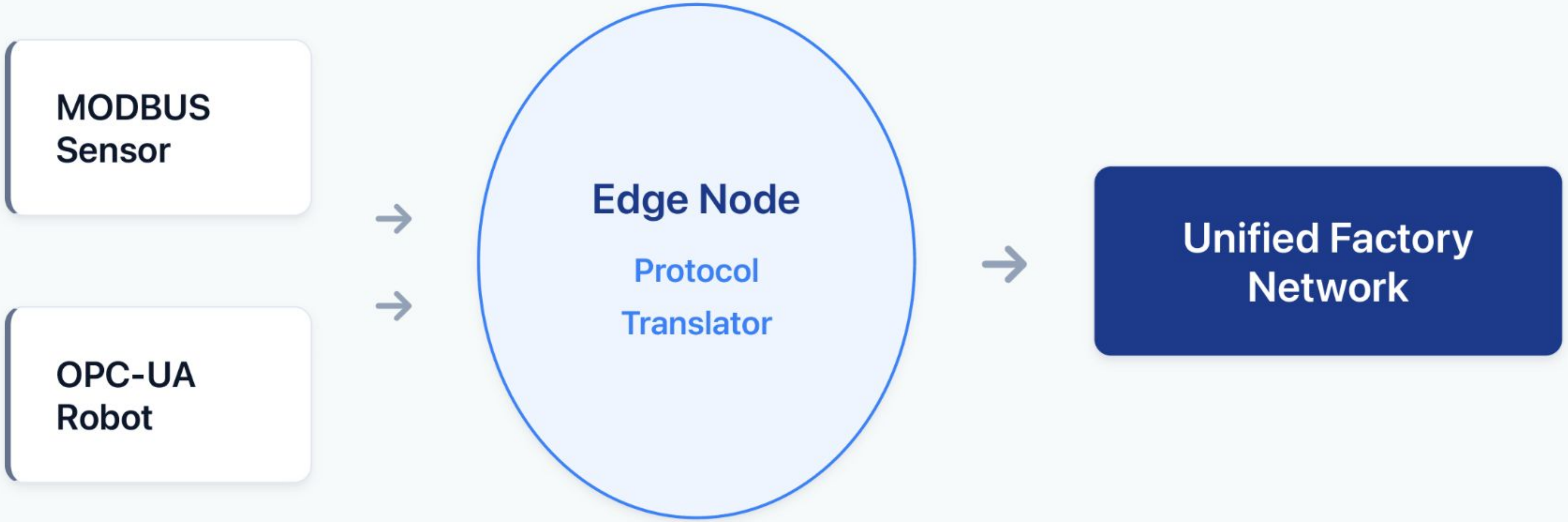
↑ **Machines generate raw data** ↑

🏭 Industrial Devices

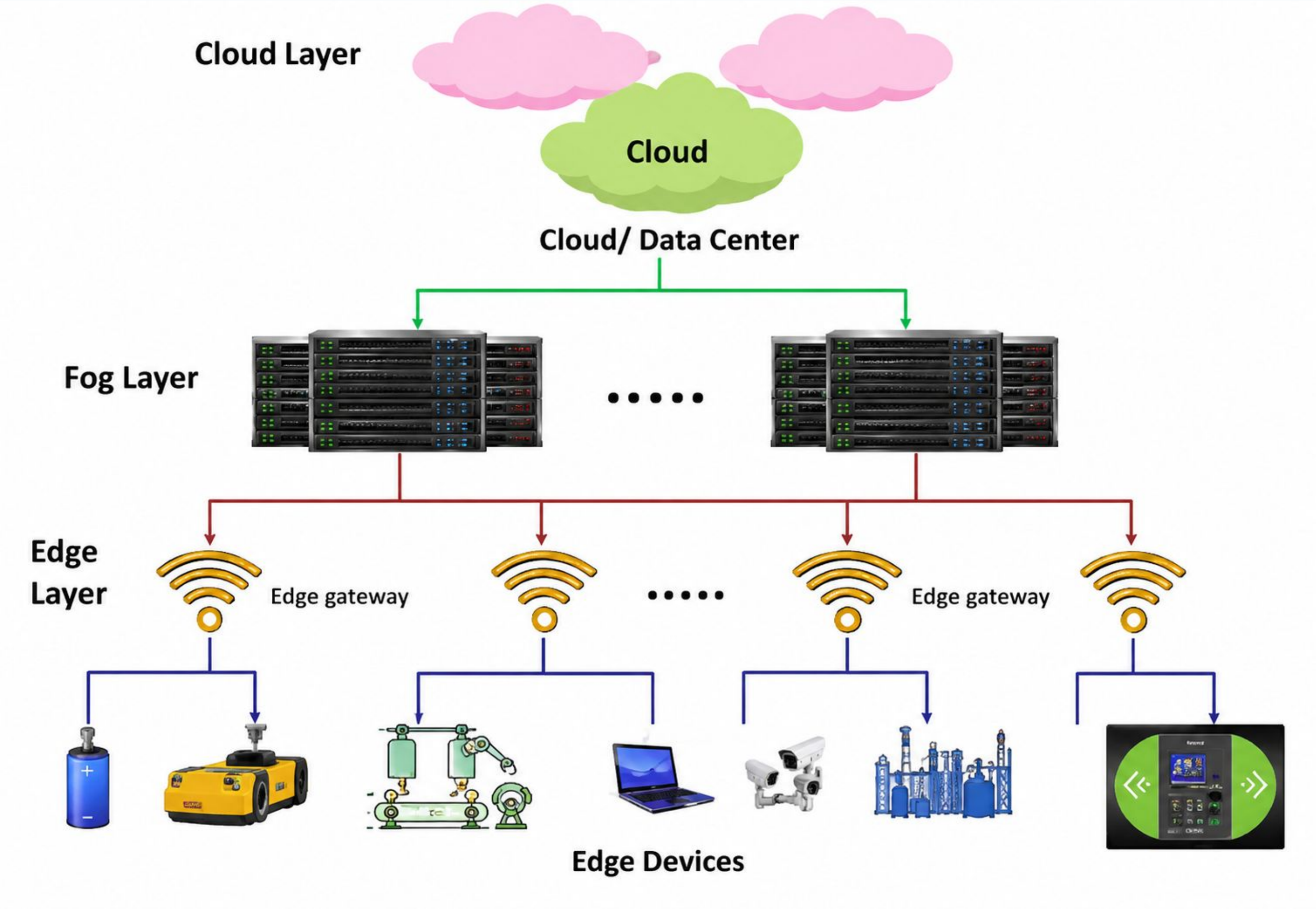
Sensors, PLCs, Cameras, Robotic Arms

Edge Node = Factory Translator

Different languages ➔ Edge translates ➔ Everyone communicates



Architecture of Edge Computing



Cloud layer	Long-term storage, dashboards, model training, ERP/MES reporting
Fog layer	Buffers and aggregates data between edge and cloud
Edge layer	Local processing, AI/fault detection, protocol translation
Edge devices	Sensors, cameras, PLCs, robots and IoT modules

Cloud vs. Edge Computing

	Cloud Computing	Edge Computing
Processing Location	Central cloud servers	**Near the device/user**
Latency	Higher	**Lower**
Bandwidth Use	High — all data to cloud	**Low — only summaries sent**
Storage Capacity	Very high (unlimited)	**Limited (local storage)**
Reliability	Requires internet connection	**Works offline**
Best Use Case	Big data · ML training · ERP	**Real-time control · Safety**

Latency test Setup

Latency Definition

Tested in two environments: Paderborn home Wi-Fi (stable, private) · HSHL central Wi-Fi (shared, congested)

Client device

Mobile phone

Generated sensor-like data

Edge server

Ubuntu laptop

Processed the request locally

Cloud target

Internet server

External online server

Measured latency

Request sent

Response received

Time from sending a request until the response is received.

Key Benefits

Low Latency

Processing near the source cuts response times to milliseconds — essential for autonomous vehicles, robotics, and emergency healthcare.

Reduced Bandwidth

Only key results (alarms, status updates) travel to the cloud. Raw data stays local, saving network capacity.

Improved Privacy

Sensitive data (e.g. patient records) stays on-premise. Only necessary summaries reach the cloud.

Better Reliability

Edge systems continue operating when cloud connectivity is poor or lost — critical for remote factories.

Application Areas

Internet of Things

Smart homes, factories & agriculture process sensor data locally for instant reactions.

Autonomous Vehicles

Cameras and LIDAR data processed on-vehicle — cloud round-trips are too slow for safety.

Healthcare

Wearables and patient monitors trigger local emergency alerts within milliseconds.

Industrial Automation

Fault detection, quality control, and real-time machine monitoring on the factory floor.

Smart Cities

Traffic sensors and cameras analysed locally to reduce congestion and improve safety.

Key Insight

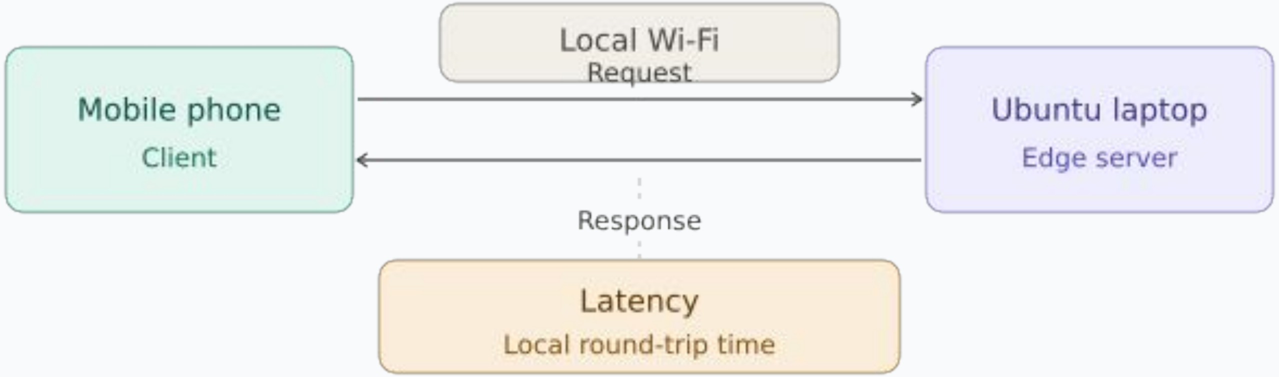
In every domain, the unifying theme is the same: decisions that matter happen in real time, and real time means milliseconds — not the 500 ms+ of a cloud round-trip.

Edge computing bridges the gap between raw sensor data and timely, local intelligence.

Only “HSHL” Wifi USER

Scan Me

Edge Latency Test



Test Procedure

Server: Ubuntu 22.04 running Python Flask
Client:
Mobile phone ⟶ http://192.168.178.78:5000
Terminal log:

```
POST /process HTTP/1.1" 200
192.168.178.117 → .78:5000
```

Status 200 OK confirms successful edge communication.

Stable local network

95 ms

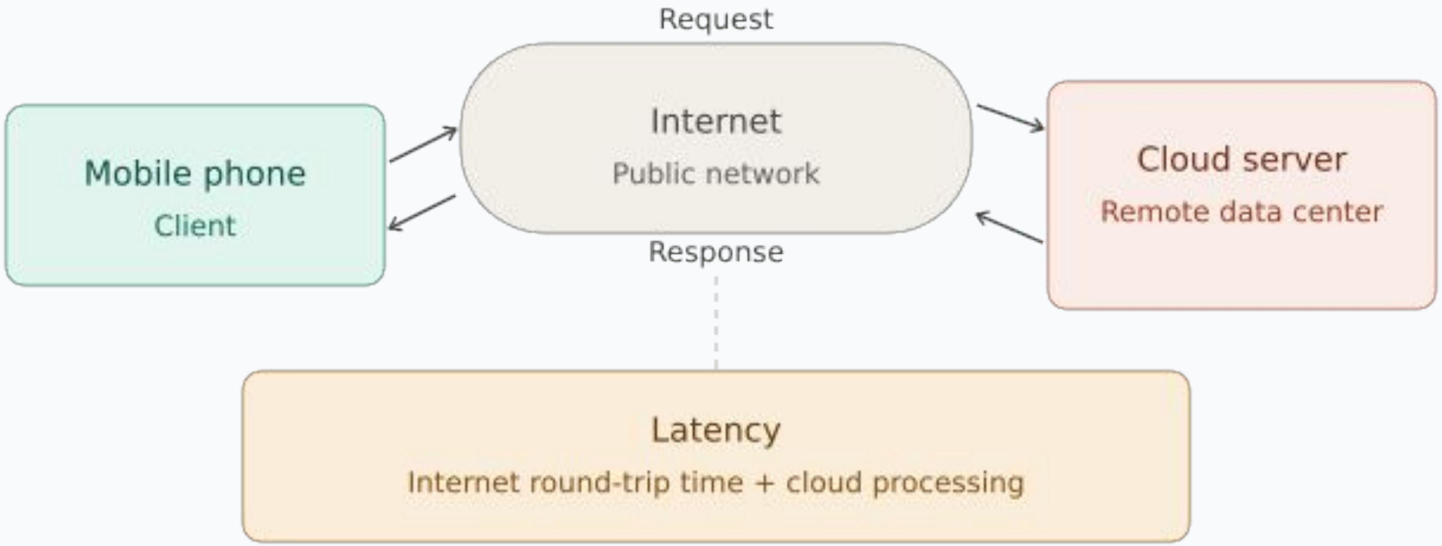
Paderborn home Wi-Fi

Shared network

422 ms

HSHL central Wi-Fi

Cloud Latency Test



Test Procedure

Method: HTTP request via public internet
Target:
https://httpbin.org/get
Command:

```
curl -w "%{time_total}s"
https://httpbin.org/get
```

Result: 0.510079 s = **510 ms** internet round-trip.

Measured cloud latency

510 ms

Paderborn home Wi-Fi

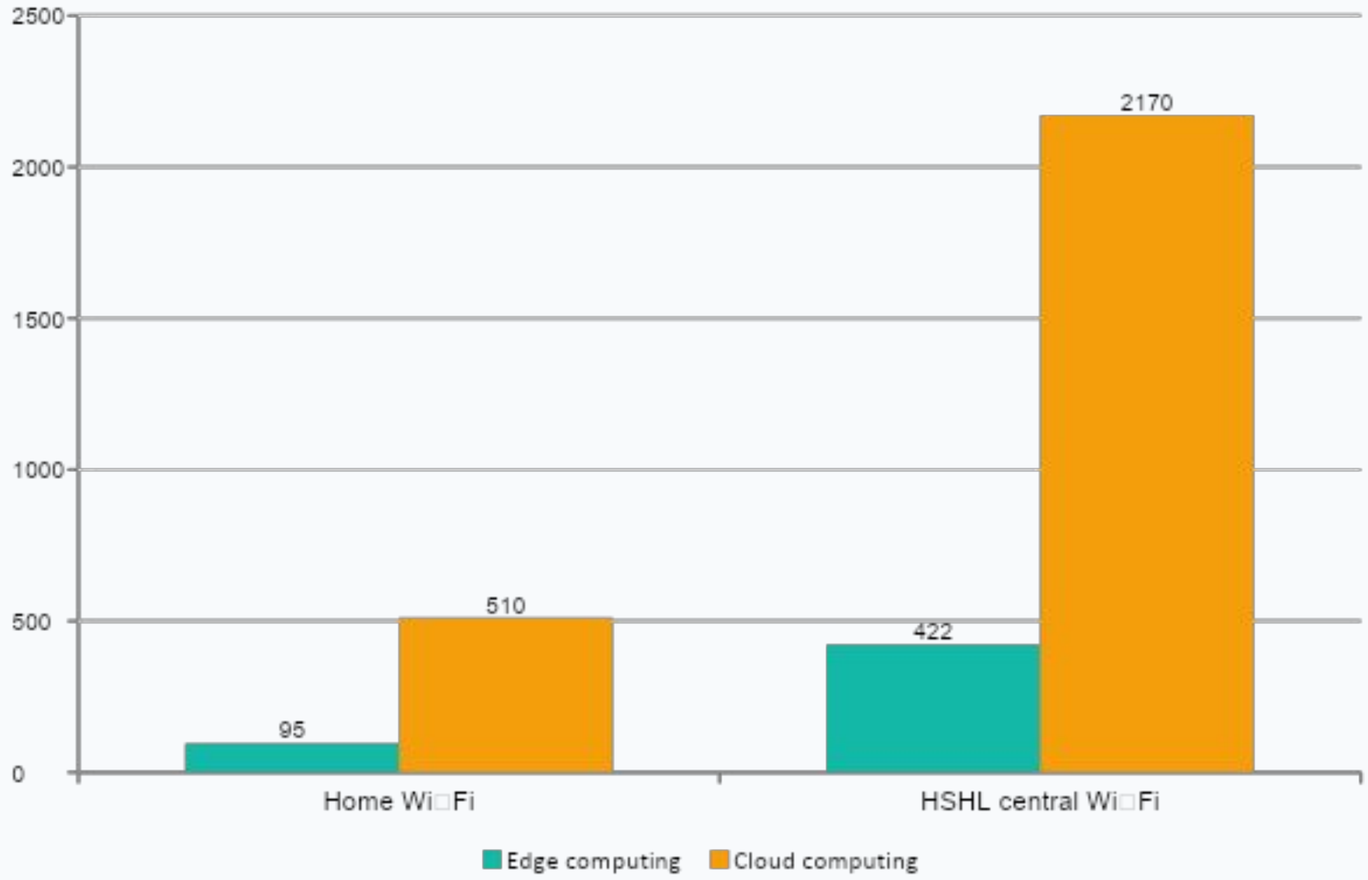
Worst shared network result

2170 ms

HSHL central Wi-Fi

Measured Results: Edge vs Cloud

Latency in milliseconds (lower is better)



Home Wi-Fi

95 vs 510 ms

Cloud took about 5.4× longer

HSHL Wi-Fi

422 vs 2170 ms

Cloud took about 5.1× longer

Latency Formulation

$$C = X / L + D / R$$

C = latency

X = distance • L = bandwidth • D = data rate • R = computing resources

Edge path

short X

Cloud path

large internet X

What increases delay?

- Edge reduces the distance X because processing is local.
- A shared network lowers available bandwidth L.
- Cloud routes are longer, so strict millisecond response is harder.

Edge vs. Cloud: Full Comparison

Criterion	Edge Computing	Cloud Computing
Latency (stable network)	**95 ms**	**510 ms**
Latency (shared network)	**422 ms**	**2,170 ms**
Speed advantage	**5.1× – 5.4× faster in every test**	
Offline operation	✓ **Yes**	✗ **No**
Suitable for real-time	✓ **Yes**	Limited
Network dependency	Local network	Internet required
Storage & analysis scale	Limited	✓ **High**

Edge is best for fast local actions. Cloud is best for heavy storage, training, reporting, and long-term analytics.

From Experiment to Industrial System

Experiment	Industrial Equivalent
Mobile Phone	Sensor / PLC / Robot
Wi-Fi	Industrial Ethernet / OPC UA / 5G
Ubuntu Laptop	Edge Gateway
Flask Application	Edge Analytics
Latency Measurement	Response Time

Real Industrial Applications

PCB Assembly

Cameras inspect boards instantly. Detects missing components and rejects boards before moving down the line.

CNC Machining

Monitors vibration and motor current. Detects tool wear locally to prevent damage to expensive workpieces.

Robotic Welding

Evaluates voltage and weld times in milliseconds. Stops defective welds immediately.

Pharma Packaging

Verifies fill levels and label positions. Removes faulty or underfilled products directly on the spot.

Challenges & Limitations

Security Risks

Edge devices are distributed across many sites, making centralised protection difficult. A compromised device can impact the whole system.

Limited Resources

Edge hardware has less compute, memory, and storage than cloud servers. Applications must be carefully optimised.

Management Complexity

Coordinating software updates, monitoring, and fault detection across hundreds of edge nodes is a significant operational challenge.

Data Consistency

Processing data at multiple sites makes it harder to maintain a single consistent view between edge and cloud systems.

Conclusion

- Edge computing is a key technology for modern distributed systems, not a replacement for the cloud, but a complement.
- The edge handles latency-sensitive local tasks (fault detection, safety response, quality control).
- The cloud handles large-scale storage, model training, and long-term analytics.
- Experimental results confirm edge processing is 5.1–5.4× faster across all network conditions tested.
- Edge is best for fast local actions. Cloud is best for heavy storage, training, reporting, and long-term analytics.

Thank you · Questions welcome